Technology Stack

Date	21 October 2026
Team ID	VAI-AI-2026
Project Name	VoyageAI — Personalized AI Travel Itinerary Planner
Maximum Marks	2 Marks

Technology Stack Details & Justifications:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs used to build the solution. A well-chosen technology stack ensures that the application is scalable, maintainable, and aligned with project requirements.

S.No	Architecture Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	React 19 + Tailwind CSS v4 + Vite + Zustand	React 19 provides state-of-the-art rendering performance; Tailwind v4 delivers responsive utility styling; Zustand provides lightweight global state management without Redux boilerplate.
2	**Backend / Server-Side**	FastAPI (Python 3.12) + Uvicorn + Pydantic v2	High-throughput asynchronous event loop (async/await), automated OpenAPI/Swagger interactive docs, and strict schema validation.
3	**Database / Data Storage**	Supabase PostgreSQL 15 (SQLAlchemy 2.0 + asyncpg)	ACID relational integrity, transaction pooling on Port 6543 (statement_cache_size=0), foreign key cascading, and database-level Row-Level Security (RLS).
4	**Cloud / Hosting / Deployment**	Vercel (Frontend CDN) + Render.com (Backend Web Service)	Vercel hosts the React SPA on global Edge servers with SPA routing (vercel.json); Render hosts the containerized FastAPI web service with continuous monitoring.
5	**Version Control & CI/CD**	Git + GitHub Repository (main branch)	Distributed version control with automated webhook deployments to Vercel and Render on verified code pushes.
6	**Third-Party APIs & AI Tools**	Groq Cloud LPU (openai/gpt-oss-120b) + Lucide + ReportLab	Groq Language Processing Units deliver ultra-low latency JSON generation; Lucide provides iconography; ReportLab generates pixel-perfect PDF submission deliverables.